TRIBHUVAN UNIVERSITY

PATAN MULTIPLE CAMPUS

PATANDHOKA, LALITPUR

LAB REPORT

SOFTWARE ENGINEERING (BIT 302)

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DATE: 2082-12-05

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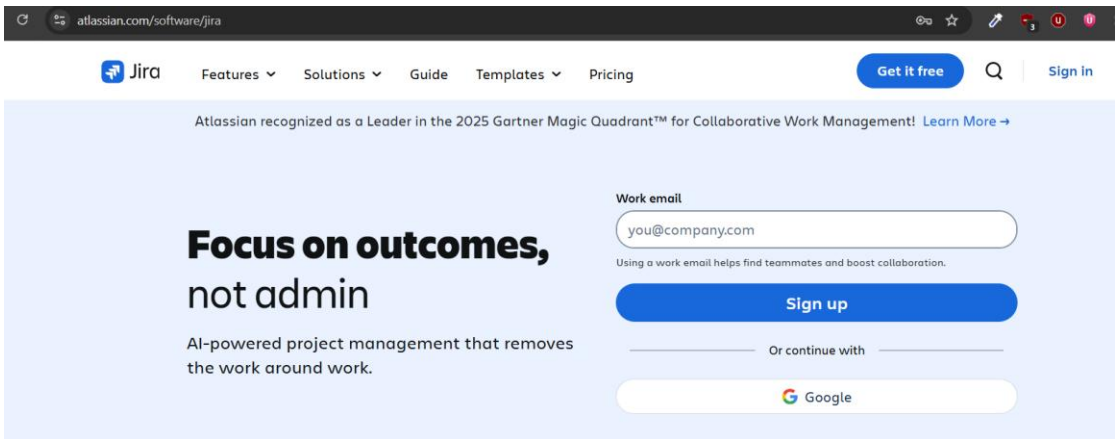
LAB-4: FAMILIARIZATION OF JIRA

THEORY:

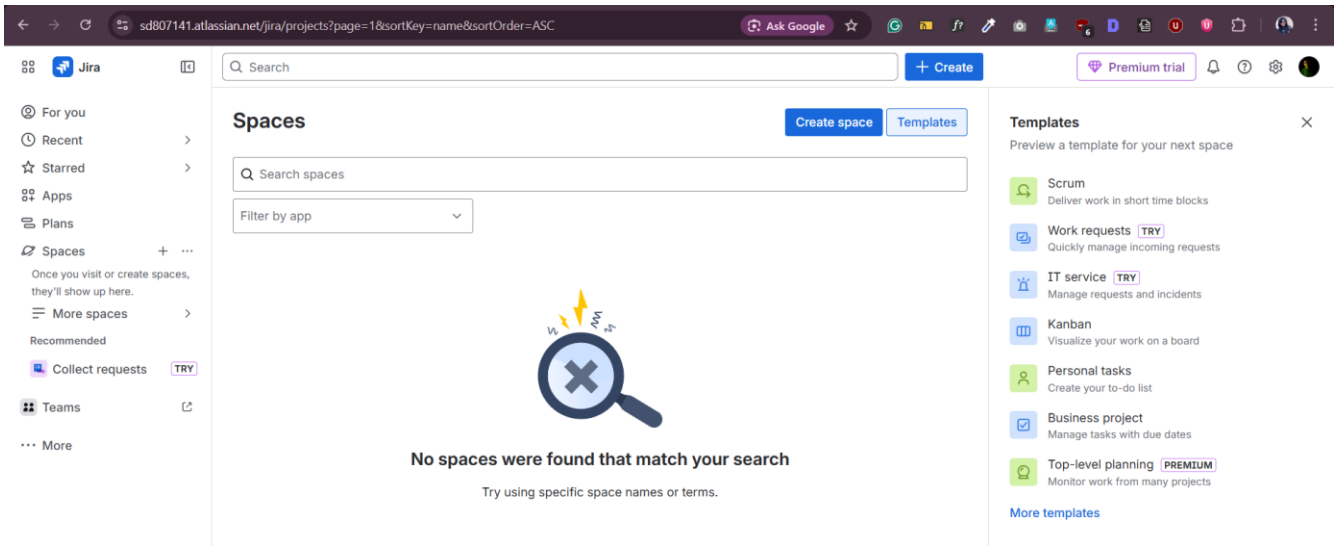
Jira is a project management and issue-tracking tool developed by Atlassian. It is commonly used in software development to manage tasks, bugs, and project activities in an organized way. In Jira, work is divided into projects, and each project contains issues such as tasks, user stories, or bugs. These issues move through different stages like To Do, In Progress, and Done, which helps in tracking the progress of work. Jira also supports Agile methodologies like Scrum and Kanban. It provides features such as boards, backlogs, and sprint planning, which make it easier for teams to collaborate and complete work efficiently.

GETTING STARTED:

Jira can be accessed through web at https://www.atlassian.com/software/jira.

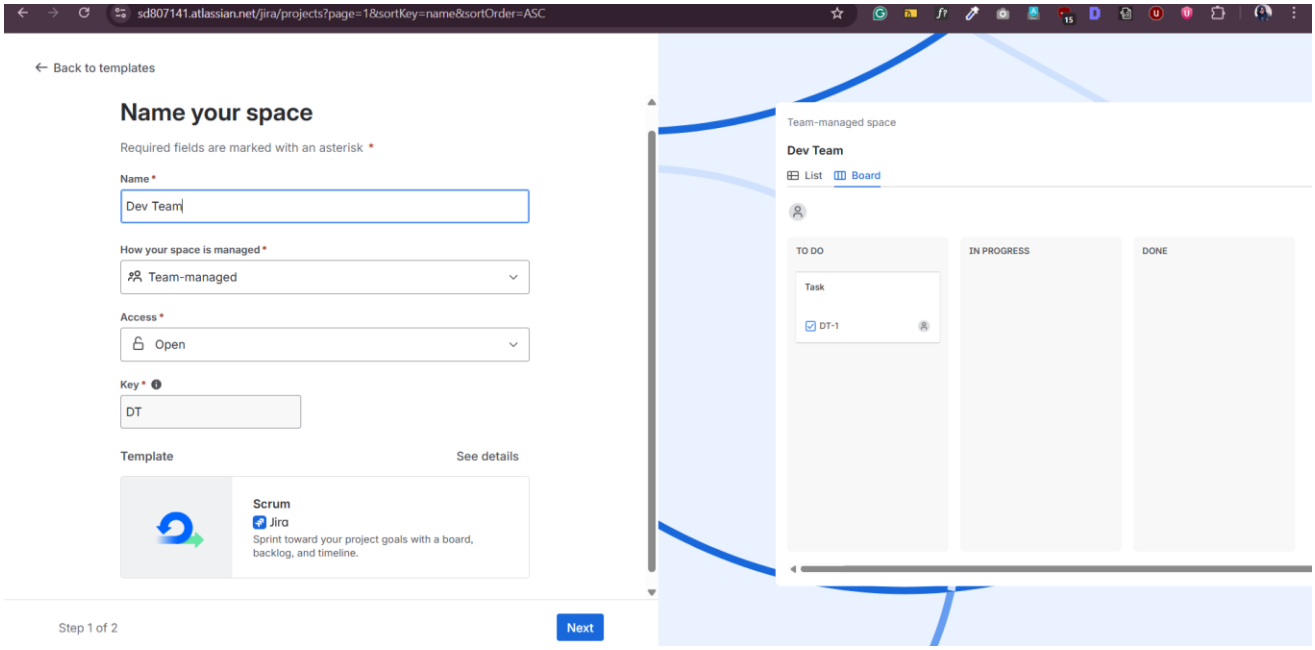


We must create an account or login to use Jira. After signing up, there is a brief onboarding process after which we will be redirected to the dashboard.

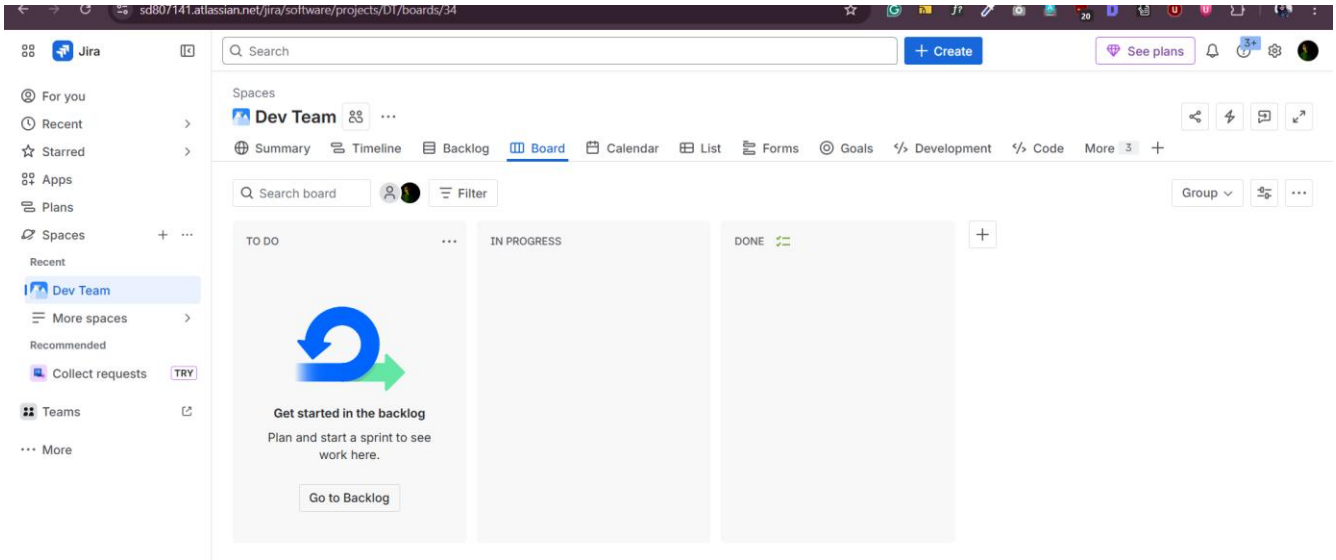


CREATING SPACE:

From right sidebar we selected Scrum template and selected team managed space.

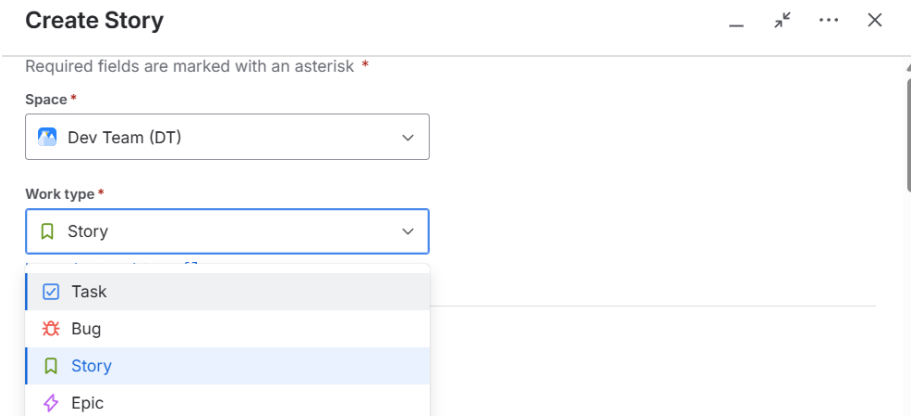


We were redirected to the newly created space page.

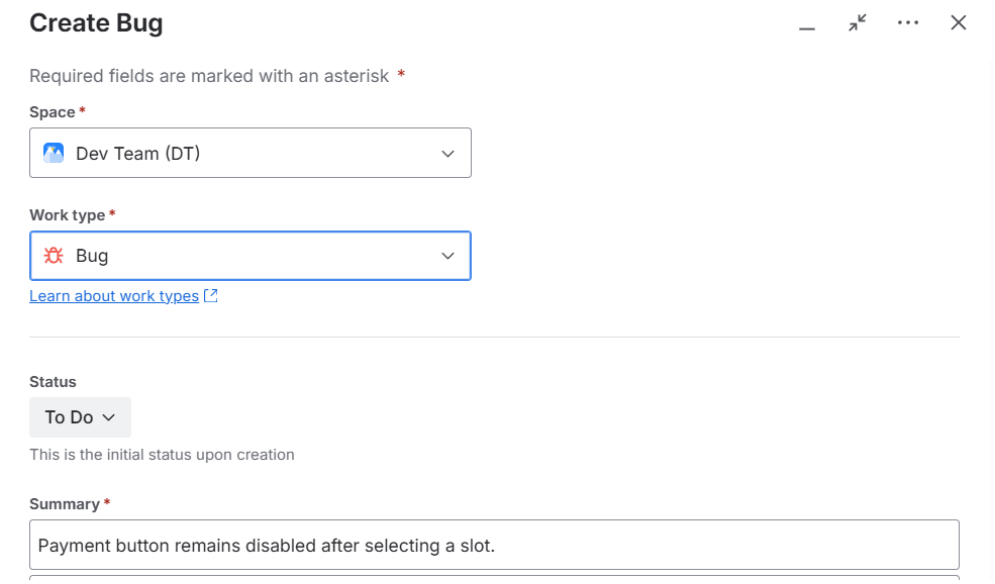


CREATING ISSUES:

We navigated to backlog section, clicked on create button and created user stories. Backlog is a section where we dump all the requirements or tasks of the system prioritized in order.



We also created a bug by selecting work type bug.



There are different issue types in Jira such as story, bug, task, work, epic etc.

1. Epic

An Epic represents a large body of work that can be broken down into several smaller tasks. It usually takes multiple weeks or "Sprints" to complete.

- **Example:** “Online Payment Module” or “User Account Management.”
- **Role in Report:** To show high-level project planning.

2. Story

Also known as a User Story, this is the most common work type. It describes a feature from the perspective of the person using the software.

- **Format:** As a [user], I want to [action], so that [benefit].
- **Example:** As a player, I want to see available time slots so I can book a pitch.
- **Role in Report:** This demonstrates Requirement Engineering.

3. Task

A Task is used for technical work that doesn't necessarily have a "user story" format. These are things the developers just need to get done.

- **Example:** Set up Database Server
- **Role in Report:** Shows the technical work required.

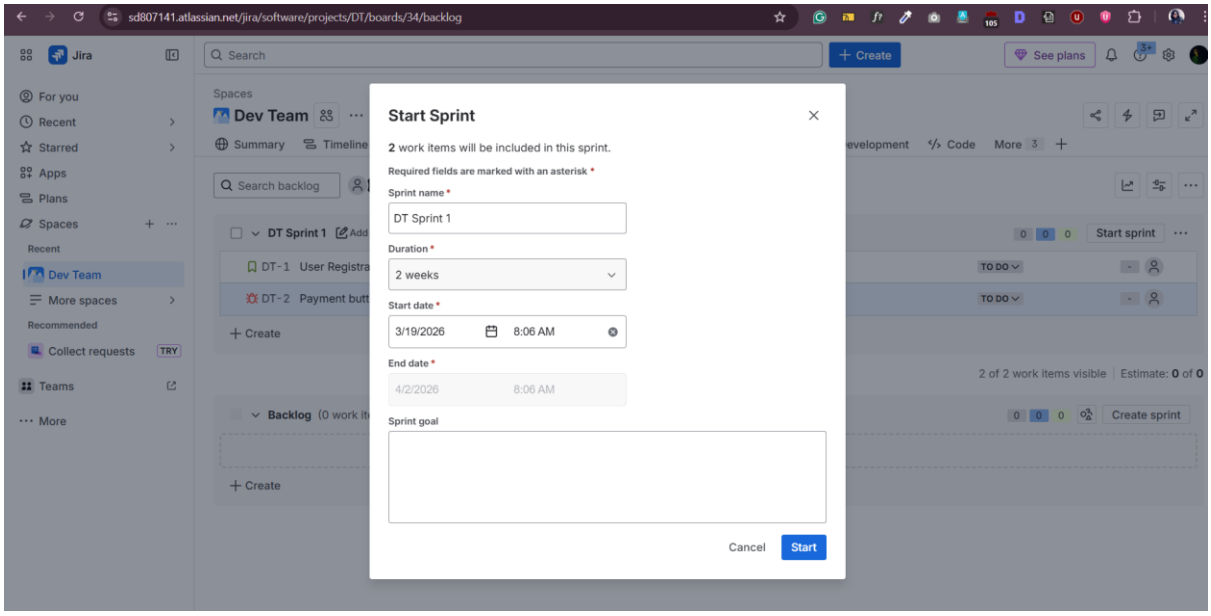
4. Bug

A Bug is an issue that describes a failure or a flaw in the system that needs to be fixed.

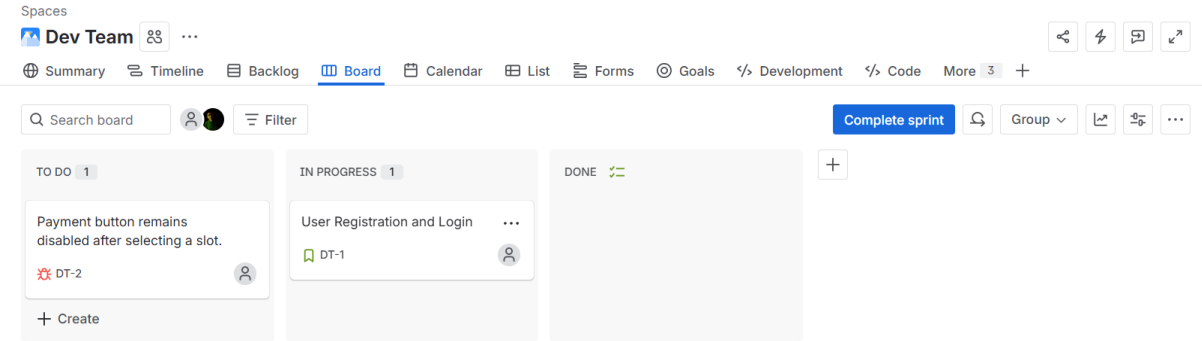
- **Example:** The app crashes when a user tries to upload a profile picture larger than 5MB.
- **Role in Report:** Demonstrates Quality Assurance and maintenance.

STARTING SPRINT:

We dragged the issues from backlog section to sprint section and clicked on start sprint button to start sprint.

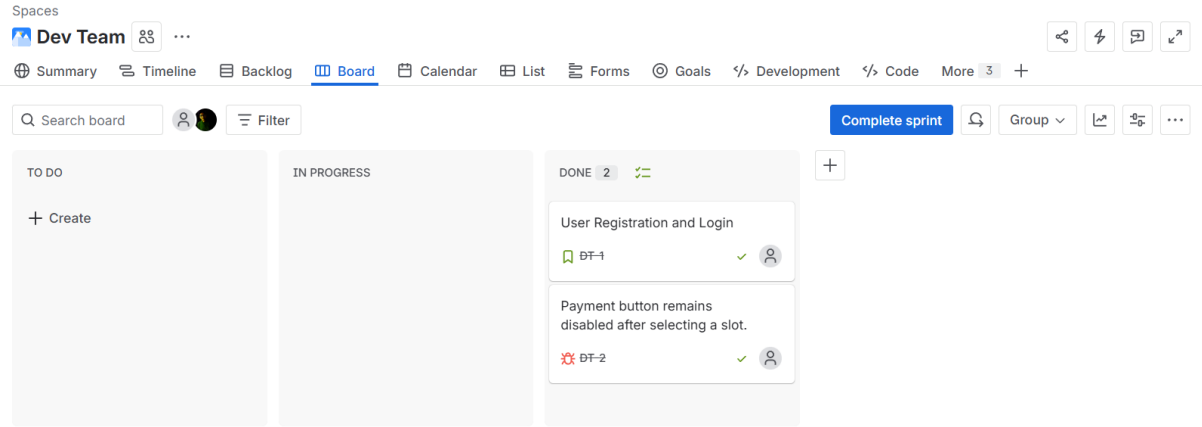


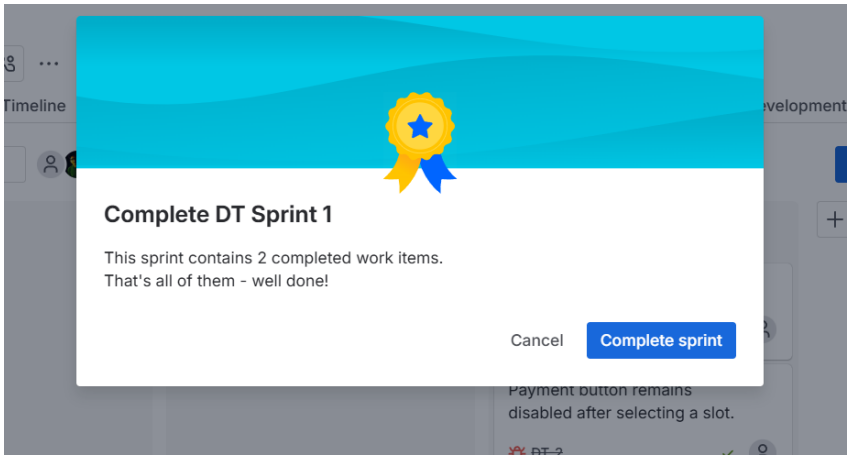
After starting sprint, we navigated to the board tab. There were three boards, to do, in progress, and done. Here we can drag and drop issue into different boards to track progress.



COMPLETING SPRINT:

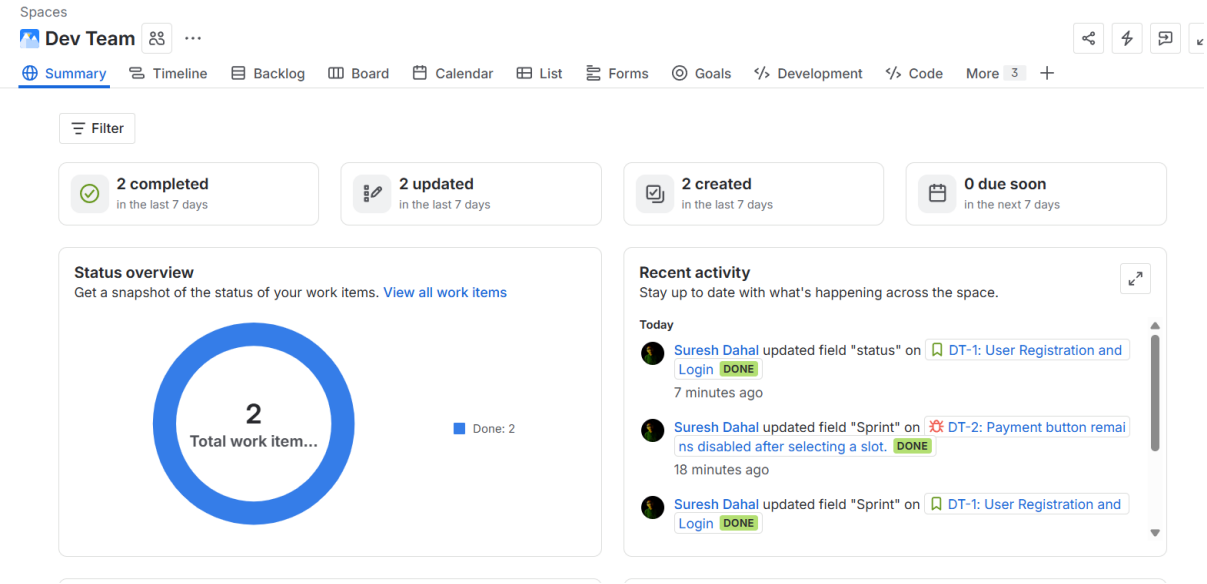
After completing the tasks, we can complete the sprint by clicking the complete sprint button. Which removes all the issues related to that sprint from backlog and board section.





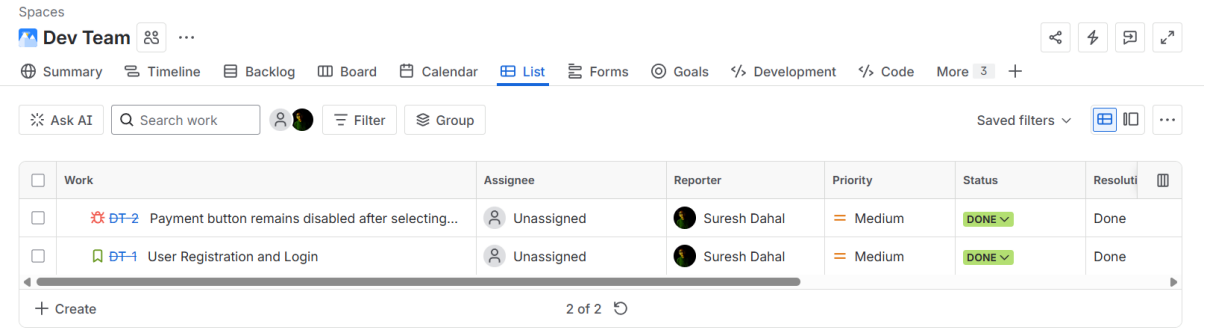
VIEWING SUMMARY:

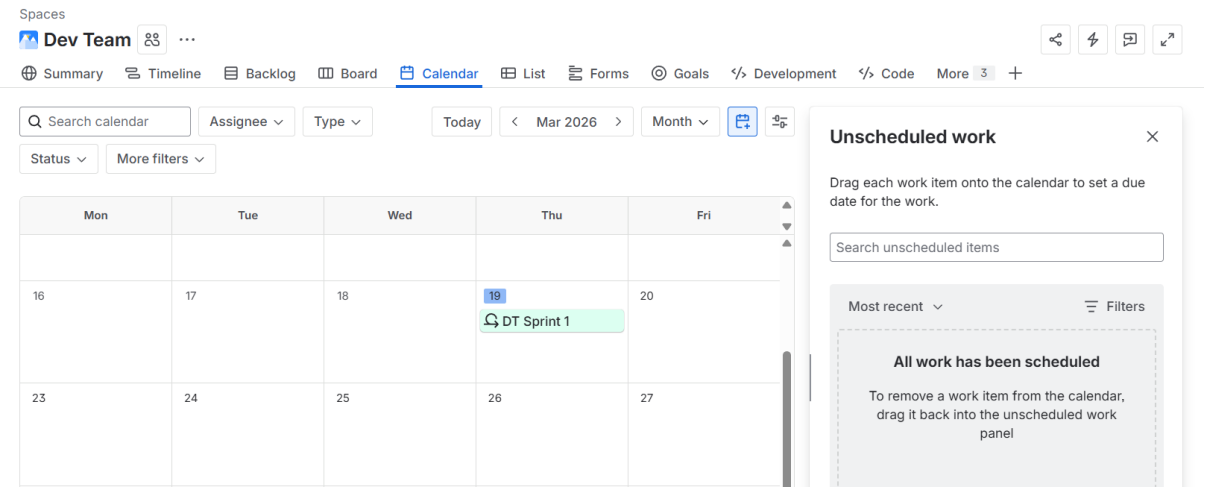
We can navigate to summary tab to view overall summary of that space. It shows number of tasks completed, created, and tasks whose deadlines are approaching.



DIFFERENT VIEWS:

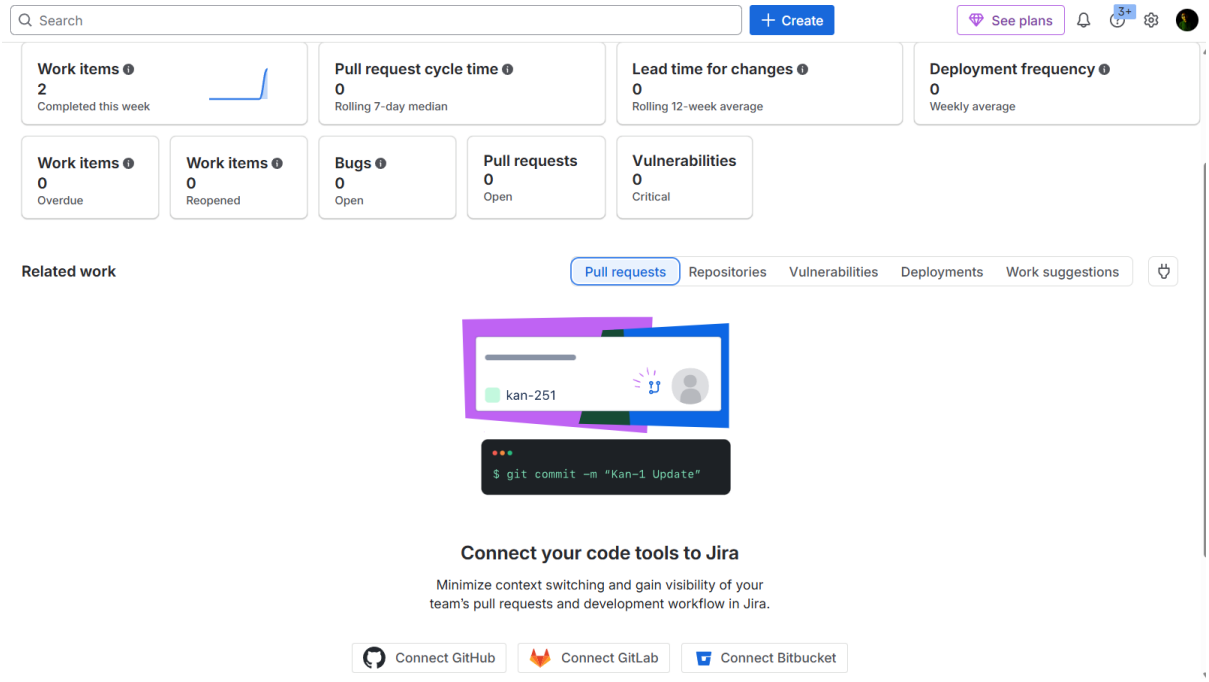
Jira gives different views of issues. Like calendar view, board view, list view etc.



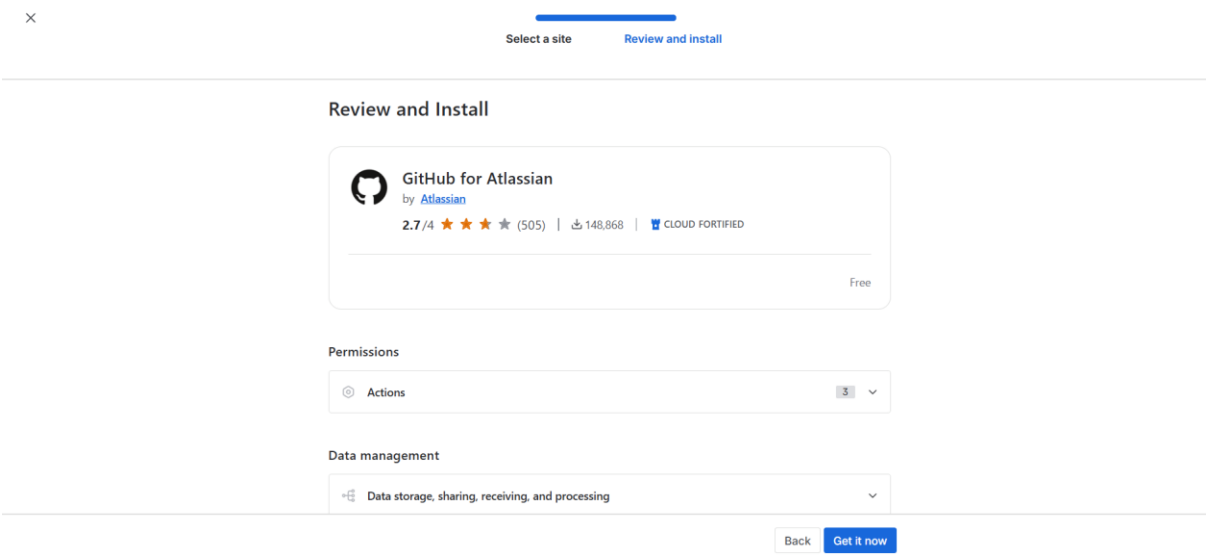


INTEGRATIONS:

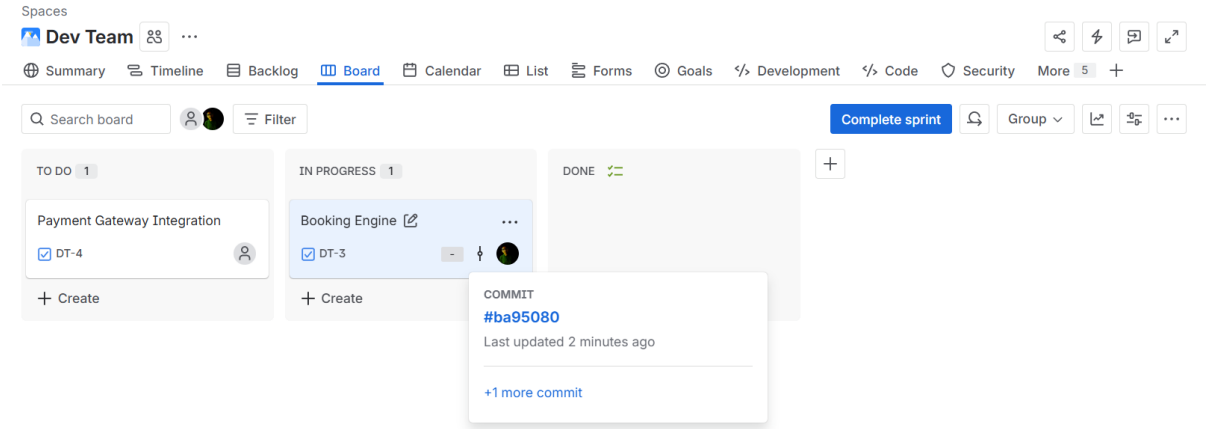
We can navigate to development tab and integrate source control tools such as GitHub and GitLab. It allows Jira to track changes automatically.



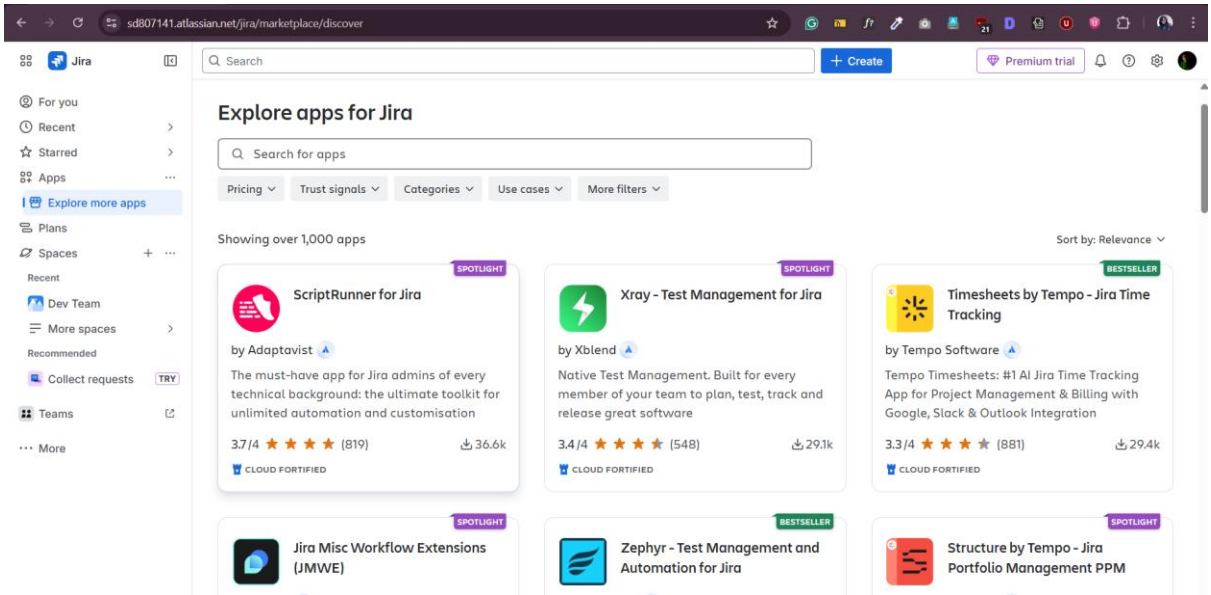
We connected Jira with GitHub.



After connecting GitHub with Jira, we can see commit directly in it.



We can also integrate other apps. There are plenty of configurable apps.



CONCLUSION:

Hence, by doing this lab, we became familiar with Jira. We learned the basic features of Jira and how we can use this tool to manage tasks, bugs, and project activities in an organized way.